Dasari Naga Raju

Project Research Assistant at Medical College of Wisconsin, USA

Andhra Pradesh, India | raajuuu1998@gmail.com | **Project TARAA** | | | |

Research Interests

My research interests center on biologically grounded representation learning for molecular biomarker and phenotype prediction from H&E whole-slide images, using pathology foundation models and weakly supervised learning. I am particularly interested in improving generalization under cross-cancer, cross-site, and combined distribution shifts, while distinguishing transferable biological patterns from cancer-specific signals.

Education

International Institute of Information Technology Bangalore *Sept 2021 – July 2024*
M.S. by Research in Information Technology CGPA: 3.16 / 4.0
Research Area: Computational Pathology | 59/75 credits dedicated to supervised thesis research

Jawaharlal Nehru Technological University Gurajada, Vizianagaram *July 2016 – Sept 2020*
B.Tech. in Electronics and Communication Engineering CGPA: 7.6 / 10

M.S. Thesis

DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology *Published at MICCAI 2026 Workshop on BrainWorks* [Paper]

- Demonstrated that explicitly encoding nuclear spatial patterns provides discriminative information beyond tissue appearance and general-purpose pathology foundation models in data-limited settings; DeepNucleiNet achieved a mean **F1-score of 0.942**, compared with **0.884** for UNI2-h and **0.730** for Virchow2 under strict inter-patient evaluation.
- Evaluated cross-tumour generalization without retraining, achieving an **F1-score of 0.943** on TCGA-GBM, **0.843** on DeepHisto, and **0.635** on TiGER. The model produced no necrosis predictions across **85,632** TCGA-LGG negative-control tiles.

Independent Computational Pathology Research

Tissue-Aware Representation Adaptation Across Cancers (Project TARAA) *Jan 2025 – Present*

Through Project TARAA, I independently design, implement, and evaluate methods that incorporate spatial tissue architecture and biological domain knowledge into pathology foundation-model representations. I assess their value for molecular biomarker and rare joint molecular phenotype prediction, with attention to cross-cohort robustness and potential clinical utility.

Publications and Preprints

1. **Dasari Naga Raju**, "TIRA: Tumor Immune Representation Adaptation for Zero-Shot Cross-Cancer MSI and TMB Prediction." *Preprint, **arXiv**, 2026.* **[Paper] [Code]**
2. **Dasari Naga Raju**, "Conserved Immune Topology Improves Pathology Foundation Model Generalization for Cross-Cancer MSI-H Prediction." *Published at **ECCV 2026 Workshop** on Medical Foundation Models and Benchmarks (MedFMB).* **[Paper] [Code]**
3. **Dasari Naga Raju**, Tripti Bameta, "State-Aware Interaction MIL for Rare Joint Molecular Phenotype Prediction in Colorectal Cancer and Lung Adenocarcinoma." *Accepted at **NeurIPS 2026 Workshop** on AISC: Cancer Pathology Foundation Models.* **[Paper] [Code]**
4. **Dasari Naga Raju**, "What Do Pathology Foundation Models Learn Beyond Gleason Grade in Prostate Cancer?" *Published at **MICCAI 2026 Workshop** on Computational Pathology and Multimodal Data (COMPAYL).* **[Paper] [Code]**
5. **Dasari Naga Raju**, T.K. Srikanth, Shilpa Rao, Ramesh Kestur, Anita Mahadevan, "DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology." *Published at **MICCAI 2026 Workshop** on BrainWorks.* **[Paper] [Code]**
6. **Dasari Naga Raju**, V.K. Kancharla, P. Kaveti, "Surgical Foundation Model Ensembles for Predicting Annotator Disagreement in Critical View of Safety Assessments." *Published at **MICCAI 2026 Workshop** on Clinical-driven Robotics and Embodied AI Technology (CREATE).* **[Paper]**
7. V.K. Kancharla, **Dasari Naga Raju**, P. Kaveti, "Self-Supervised Denoising Reconstruction Outperforms Foundation Models for Capsule Endoscopy Abnormality Classification." *Published at **MICCAI 2026 Workshop** on AI for Safe Surgery (SafeSurg).* **[Paper]**

Research Experience

Research Intern, Computational Pathology Laboratory *Jul 2026 – Present*

Advanced Centre for Treatment, Research and Education in Cancer (ACTREC), Tata Memorial Centre, India

Research Outcome: Accepted at the NeurIPS 2026 Cancer Pathology Foundation Models Workshop. [Paper]

- Developed a weakly supervised multiple-instance learning framework for predicting rare joint molecular phenotypes from H&E whole-slide images using pathology foundation models.
- Designed biomarker-specific representation learning with explicit interaction modeling and structured molecular-state supervision to distinguish joint-positive, single-positive, and double-negative phenotypes.
- Conducted comparative evaluations and component ablations across colorectal and lung adenocarcinoma cohorts using UNI2-h and CONCH, including analysis of molecular-state discrimination and spatial attention patterns.

Project Research Assistant, Medical College of Wisconsin, USA *Feb 2025 – Present*

- Developed an EEG de-identification tool to support privacy-preserving preparation and organization of heterogeneous clinical recordings for multi-site research.
- Designing EEG preprocessing strategies to accommodate variations in channel counts, electrode configurations, and acquisition protocols across clinical sites, reducing reliance on fixed electrode montages.
- Investigating the use of EEG foundation models and clinically informed signal patterns for focal and generalized seizure detection, with a focus on reducing model complexity, improving cross-site generalization, and enabling efficient deployment in clinical settings for patients aged 10 years and older.

Research Associate, E-Healthcare Research Centre (EHRC), IIIT Bangalore *Aug 2024 – Dec 2025*

- Extended DeepNucleiNet research through sensitivity analyses across varying proportions of necrotic tissue within image tiles, characterizing detection limitations in partially necrotic regions.
- Conducted region-wise and areal analyses across 15 pathologist-annotated necrotic regions, examining the effects of region size, tissue coverage, and interior versus boundary tiles on necrosis localization.
- Investigated the sensitivity of necrosis detection to WSI tile-grid positioning using five spatially shifted extraction grids, combining predicted regions to reduce dependence on grid alignment and mitigate missed detections.

Research Project Collaborator, Neuropathology Lab *Sept 2021 – Aug 2023*

National Institute of Mental Health and Neurosciences (NIMHANS), Bengaluru

- Processed and curated gigapixel H&E whole-slide images of meningioma, glioblastoma, and low-grade glioma for computational pathology research.
- Collaborated with neuropathologists at the NIMHANS Neuropathology Laboratory on histological annotations, biologically informed feature design, and validation of necrosis detection methods.
- Developed software tools to streamline WSI annotation, automate tile-level label generation, and support pathology dataset preparation.

Research Associate, Mirai Research Innovation Institute, Japan *Oct 2020 – Aug 2021*

- Developed a CNN-based visual inspection pipeline using ResNet50 and transfer learning for detecting missing electronic components in PCB imagery, improving classification accuracy from **82% to 94%**.

Technical Skills

Digital Pathology	Gigapixel WSI processing, OpenSlide, QuPath, ImageScope, StarDist
Tissue & Spatial Analysis	Nuclear segmentation and morphology, spatial tissue organisation, cell phenotyping, tumour microenvironment analysis
Foundation Models & MIL	UNI2-h, Virchow2, CONCH, weakly supervised learning, multiple-instance learning
Deep Learning & Vision	PyTorch, TensorFlow, Vision Transformers, visual representation learning, domain adaptation
Biomedical Signal Processing	Clinical EEG preprocessing, MNE-Python, seizure detection